**Actual Salinometer Percent: 3.5%**

- 1 point if between 0-10%
- 3 points if between 1-6%
- 3 points if between 2-5%
- 2 points if between 3-4%
- 1 point if 3.5% ←add up for total point amount

**MCQ (1 point each):**

1. A **B** C D
2. A **B** C D
3. A B **C** D
4. A **B** C D
5. A B **C** D
6. A **B** C D
7. **A** B C D
8. A **B** C D
9. **A** B C D
10. A **B** C D
11. A **B** C D
12. A **B** C D
13. A **B** C D
14. A B C **D**
15. **A** B C D
16. A **B** C D
17. A **B** C D
18. **A** B C D
19. A **B** C D
20. A **B** C D
21. A **B** C D
22. A **B** C D
23. A **B** C D
24. **A** B C D

25. **A** B C D
26. A B **C** D
27. A **B** C D
28. A **B** C D
29. **A** B C D
30. A B **C** D

**Short Answer (3 points each):**

- **1 pt for identifying correct class or tolerance level**
- **1 pt for explaining what that means for water quality**
- **1 pt for reasoning (oxygen/pollution connection)**

1. Mayflies, caddisflies, and stoneflies
    a. Class 1 – Pollution Sensitive
    b. Their presence means clean, well-oxygenated water. They die quickly if pollution or low oxygen occurs, so they indicate excellent water quality.

2. Leeches, midges, and flatworms
    a. Class 3 – Moderately Tolerant
    b. Their dominance suggests fair to poor water quality. They can survive low oxygen or moderate pollution, showing that conditions are not ideal for sensitive species.

3. Diversity as an indicator
    a. A stream with many species from Classes 1–2 has good water quality.
    b. If mostly Class 4–5 species, the water is polluted or low in oxygen.
    c. High diversity across sensitivity levels = healthier ecosystem.

4. Aquatic sowbugs and scuds
    a. Class 2 – Moderately Sensitive
    b. Indicate fair/good quality water with some organic matter and moderate oxygen. They can tolerate slight pollution but not heavy contamination.

5. Air-breathing insects (water striders, whirligig beetles)
    a. Class 5 – Air Breathing / Pollution Tolerant
    b. They get oxygen directly from the air, so they can live in polluted or low-oxygen water, unlike mayflies that depend on dissolved oxygen.

6. Changes over time
    a. If pollution increases, Class 4–5 organisms (leeches, midges, air-breathers) increase and sensitive ones disappear.
    b. If pollution decreases and oxygen improves, Class 1–2 species return, showing recovery of water quality.

**True or False:**

1. **False**
2. **True**
3. **False**
4. **True**
5. **True**
6. **False**
7. **False**
8. **True**
9. **True**
10. **False**
11. **False**
12. **True**
13. **True**
14. **True**
15. **True**

**Nitrogen Cycle Diagram:**

1. Assimilation
2. Nitrates ($NO_3^-$)
3. Decomposers
4. Ammonium ($NH_4^+$)

**Table: Water sample results from Oak Creek.**

1. Site 3 (downstream)
2. Site 3 (downstream)
3. As nitrates and phosphates increase, dissolved oxygen decreases.
4. Site 3 (downstream)
5. Possible answers: Agricultural runoff (fertilizers containing nitrate/phosphate), Urban stormwater runoff (oil, fertilizers, detergents), Wastewater discharge (septic or treatment plant effluent)

**Speed ID:**

1. Mayfly - Pollution Sensitive (Class 1)
2. Dobsonfly - Pollution Sensitive (Class 1)
3. Purple Loosestrife - Aquatic Nuisance Plant
4. Scud - Moderately Sensitive (Class 2)
5. Giant Water Bug - Air Breathing (Class 5)
6. Zebra Mussel - Aquatic Nuisance Animal
7. Midge - Moderately Tolerant (Class 3)
8. Gilled Snail - Pollution Sensitive (Class 1)

9. Predaceous Diving Beetle - Air Breathing (Class 5)
10. Whirligig Beetle - Air Breathing (Class 5)
11. Tubifex - Pollution Tolerant (Class 4)
12. Eurasian Water Milfoil - Aquatic Nuisance Plant
13. Damselfly - Moderately Sensitive (Class 2)
14. Flatworm - Moderately Tolerant (Class 3)
15. Asian Carp - Aquatic Nuisance Animal

# Milpitas Invitational - Repaired practice edition - Answer key

### Division B | Water Quality | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

### ANSWER KEY - Review only after submitting.

**repair-case. (6 points) Authored written substitute for unavailable source material: Solve this case and explain your reasoning: A branching attached organism is photographed. Can shape distinguish coral from sponge?**

reasoning: Both groups can be branching and attached.
Seek polyp/tissue, pores, and skeleton evidence.
Keep identification broad until diagnostic features are visible. (3 points).

conclusion: Shared shape is not a unique taxonomic character. (3 points).